

PROOF OF TWO CONJECTURES OF TAVAKOLI AND MORELLI ON SCHMIDT-NUMBER WITNESSES

ANSH MISHRA AND ARYAN SENTHILKUMAR

ABSTRACT. Tavakoli and Morelli [Phys. Rev. A **110**, 062417 (2024)] proposed Schmidt-number witnesses based on the trace norm of the matrix of joint outcome probabilities of two local measurements. They conjectured tight bounds for incomplete sets of mutually unbiased bases (their Conjecture 3) and for equiangular measurements (their Conjecture 2). We prove both conjectures. For m mutually unbiased bases in $\mathbb{C}^d$ on each side and every state of Schmidt number at most r , the correlation matrix satisfies $\|Q_m\|_{\text{tr}} \leq 1 + (m-1)r/d$, and this is tight for every d, m and r . For equiangular measurements with $n > d$ outcomes, $\|P_n\|_{\text{tr}} \leq [d(d-1) + r(n-d)]/[n(n-1)]$. Both results follow from a single inequality for rank-one measurements whose frame operator fixes the identity. Its proof combines a Gram factorisation of the correlation matrix, Hölder’s inequality for Schatten norms, and a Bessel-type bound for the measurement frame. Both theorems are formalised in the Lean 4 proof assistant with the Mathlib library, using only Lean’s standard axioms.

1. INTRODUCTION

The Schmidt number of a bipartite state is the smallest r such that the state is a mixture of pure states of Schmidt rank at most r [2]. It quantifies the dimensionality of entanglement, and certifying it from few measurements is a basic task in high-dimensional quantum information; see, e.g., [3, 4]. Tavakoli and Morelli [1] proposed witnesses built from the matrix of joint outcome probabilities when Alice and Bob each measure one of several local bases or a single informationally rich measurement. The criterion is an upper bound on the trace norm of this matrix, valid for all states of Schmidt number at most r ; a larger trace norm certifies Schmidt number larger than r . For incomplete sets of mutually unbiased bases (MUBs) [5, 6] and for equiangular measurements they conjectured the tight value of this bound (Conjectures 3 and 2 of [1]).

We prove both conjectures (Theorems 2 and 3). Both are corollaries of a general inequality (Theorem 5): if the frame operator of a rank-one measurement maps the identity to a multiple of itself, with eigenvalue α , and is bounded by β on traceless operators, then states of Schmidt number at most r give correlation matrices of trace norm at most $\beta + (\alpha - \beta)r/d$ (when both parties use such measurements with the same constants). The proof has two ingredients. First, a Gram factorisation of the correlation matrix of a pure state, combined with Hölder’s inequality, bounds its trace norm by a product of two “index of coincidence” quantities, one for each party (Lemma 6). Second, a Bessel-type bound controls these quantities through the trace and the Hilbert–Schmidt norm of the modulus of the coefficient matrix (Lemmas 7 and 8). The Schmidt rank enters only through the elementary inequality $\|X\|_1^2 \leq r\|X\|_2^2$.

A natural first attempt applies the triangle inequality to the Schmidt decomposition. It gives only $\|Q_m\|_{\text{tr}} \leq r + (m-1)/d$ for Schmidt rank r , which exceeds the tight value by $(r-1)(d+1-m)/d$ (Remark 13). The factorisation above avoids this loss.

Both main theorems have been formalised in Lean 4 [8] with Mathlib [9] (Section 7).

2. SETTING AND RESULTS

Throughout, $d \geq 2$, vectors live in $\mathbb{C}^d$ with its standard basis, and $\bar{v}$ denotes the entrywise complex conjugate of a vector v . The trace norm $\|A\|_{\text{tr}} = \|A\|_1$ of a matrix is the sum of its singular values, and $\|A\|_2 = (\text{Tr } A^\dagger A)^{1/2}$ is the Hilbert–Schmidt (Frobenius) norm. For a matrix X we write $|X| = (X^\dagger X)^{1/2}$ and $|X^\dagger| = (XX^\dagger)^{1/2}$.

- Definition 1.** (1) *MUBs*. A family $\{e_a^k\}$, $k = 1, \dots, m$, $a = 1, \dots, d$, consists of m mutually unbiased bases if each $\{e_a^k\}_a$ is an orthonormal basis of $\mathbb{C}^d$ and $|\langle e_a^k | e_b^l \rangle|^2 = 1/d$ for $k \neq l$.
 (2) *Equiangular measurements (EAMs)*. Unit vectors $\psi_1, \dots, \psi_n \in \mathbb{C}^d$ with $n > d$ form an equiangular measurement if $\sum_a |\psi_a\rangle\langle\psi_a| = \frac{n}{d}\mathbb{I}$ and $|\langle\psi_a|\psi_b\rangle|^2 = c$ for $a \neq b$, where necessarily $c = c_{n,d} = \frac{n-d}{d(n-1)}$. The measurement operators are $E_a = \frac{d}{n}|\psi_a\rangle\langle\psi_a|$.
 (3) *Schmidt number*. A state ρ on $\mathbb{C}^d \otimes \mathbb{C}^d$ has Schmidt number at most r , written $\text{SN}(\rho) \leq r$, if it is a finite convex combination of projectors onto unit vectors of Schmidt rank at most r [2].
 (4) *Correlation matrices* [1]. For MUBs $\{e_a^k\}$ (Alice) and $\{f_b^l\}$ (Bob), $Q_m(\rho)$ is the real $md \times md$ matrix with entries $Q_{(k,a),(l,b)} = \langle e_a^k \otimes f_b^l | \rho | e_a^k \otimes f_b^l \rangle$. For EAMs $\{\psi_a\}$ and $\{\varphi_b\}$, $P_n(\rho)$ is the $n \times n$ matrix with entries $P_{ab} = \text{Tr}[\rho(E_a \otimes F_b)] = \frac{d^2}{n^2} \langle \psi_a \otimes \varphi_b | \rho | \psi_a \otimes \varphi_b \rangle$.

The value of $c_{n,d}$ follows by taking the trace of $(\sum_a |\psi_a\rangle\langle\psi_a|)^2 = \frac{n^2}{d^2}\mathbb{I}$.

Theorem 2 (Conjecture 3 of [1]). *Let $\{e_a^k\}$ and $\{f_b^l\}$ be two sets of $m \geq 1$ mutually unbiased bases of $\mathbb{C}^d$, and let $\text{SN}(\rho) \leq r$. Then*

$$\|Q_m(\rho)\|_{\text{tr}} \leq 1 + \frac{(m-1)r}{d}.$$

For every d , every m for which m MUBs exist in $\mathbb{C}^d$, and every $1 \leq r \leq d$, equality is attained.

Theorem 3 (Conjecture 2 of [1]). *Let $\{\psi_a\}$ and $\{\varphi_b\}$ be equiangular measurements with $n > d$ elements in $\mathbb{C}^d$, and let $\text{SN}(\rho) \leq r$. Then*

$$\|P_n(\rho)\|_{\text{tr}} \leq \frac{d(d-1) + r(n-d)}{n(n-1)}.$$

Equality is attained for $r \in \{1, d-1, d\}$ for every EAM, and for every r whenever the real span of the projectors $|\psi_a\rangle\langle\psi_a|$ contains a projector of rank r (for instance for SICs, $n = d^2$).

Both theorems are special cases of the following inequality. For a finite family $a = \{a_i\}$ of vectors in $\mathbb{C}^d$ (not necessarily normalised) and a Hermitian matrix A , write

$$I_a(A) = \sum_i \langle a_i | A | a_i \rangle^2.$$

Definition 4. A family a has *frame constants* (α, β) , with $\alpha \geq \beta \geq 0$, if for every Hermitian A

$$(1) \quad I_a(A) \leq \beta \text{Tr}(A^2) + (\alpha - \beta) \frac{(\text{Tr } A)^2}{d}.$$

Theorem 5 (general bound). *Let $a = \{a_i\}_{i \in I}$ and $b = \{b_j\}_{j \in J}$ be families in $\mathbb{C}^d$ with frame constants (α_a, β_a) and (α_b, β_b) , and let $M(\rho)$ be the $|I| \times |J|$ matrix with entries $M_{ij} = \langle a_i \otimes b_j | \rho | a_i \otimes b_j \rangle$. If $\text{SN}(\rho) \leq r$, then*

$$\|M(\rho)\|_{\text{tr}} \leq \left(\beta_a + (\alpha_a - \beta_a) \frac{r}{d} \right)^{1/2} \left(\beta_b + (\alpha_b - \beta_b) \frac{r}{d} \right)^{1/2}.$$

More precisely, for a unit vector $|\psi\rangle = \sum_{ij} X_{ij}|i\rangle|j\rangle$,

$$(2) \quad \|M(|\psi\rangle\langle\psi|)\|_{\text{tr}} \leq \left(\beta_a + (\alpha_a - \beta_a) \frac{\|X\|_1^2}{d}\right)^{1/2} \left(\beta_b + (\alpha_b - \beta_b) \frac{\|X\|_1^2}{d}\right)^{1/2}.$$

Lemmas 7 and 8 below give the frame constants

$$m \text{ MUBs: } (\alpha, \beta) = (m, 1); \quad \text{EAM vectors } \sqrt{d/n} \psi_a : (\alpha, \beta) = \left(\frac{d}{n}, \frac{d(d-1)}{n(n-1)}\right).$$

With these values Theorem 5 gives exactly the bounds of Theorems 2 and 3. For the EAM, $\beta + (\alpha - \beta) \frac{r}{d} = \frac{d(d-1)}{n(n-1)} + \frac{r}{n} - \frac{(d-1)r}{n(n-1)} = \frac{d(d-1)+r(n-d)}{n(n-1)}$.

3. PROOF OF THE GENERAL BOUND

3.1. Pure states and coefficient matrices. A vector $|\psi\rangle = \sum_{ij} X_{ij}|i\rangle|j\rangle$ of $\mathbb{C}^d \otimes \mathbb{C}^d$ is determined by its coefficient matrix $X \in \mathbb{C}^{d \times d}$. Then $\|\psi\|^2 = \|X\|_2^2$, the Schmidt coefficients of ψ are the singular values of X , and the Schmidt rank of ψ is $\text{rank } X$. For $u, v \in \mathbb{C}^d$,

$$\langle u \otimes v | \psi \rangle = \sum_{ij} \bar{u}_i \bar{v}_j X_{ij} = \langle u | X | \bar{v} \rangle,$$

so $M_{ij}(|\psi\rangle\langle\psi|) = |\langle a_i | X | \bar{b}_j \rangle|^2$.

3.2. Gram factorisation.

Lemma 6. Let $X \in \mathbb{C}^{d \times d}$ and let $\{a_i\}_{i \in I}$, $\{b_j\}_{j \in J}$ be finite families in $\mathbb{C}^d$. The matrix $M_{ij} = |\langle a_i | X | b_j \rangle|^2$ satisfies

$$\|M\|_1 \leq I_a(|X^\dagger|)^{1/2} I_b(|X|)^{1/2}.$$

Proof. Let $X = \sum_s \sigma_s |u_s\rangle\langle v_s|$ be a singular value decomposition with $\sigma_s > 0$ and orthonormal families $\{u_s\}$, $\{v_s\}$. Put

$$L = \sum_s \sigma_s^{1/2} |u_s\rangle\langle u_s|, \quad R = \sum_s \sigma_s^{1/2} |v_s\rangle\langle v_s|, \quad W = \sum_s |u_s\rangle\langle v_s|.$$

Then $LWR = X$, $L^2 = |X^\dagger|$, $R^2 = |X|$, and $W^\dagger WR = R$, because $W^\dagger W = \sum_s |v_s\rangle\langle v_s|$ is the projector onto the range of R . Let $x_i = La_i$ and $y_j = WRb_j$. Since L is Hermitian, $\langle a_i | X | b_j \rangle = \langle x_i | y_j \rangle$. With $\xi_i = x_i \otimes \bar{x}_i$ and $\eta_j = y_j \otimes \bar{y}_j$ we get $\langle \xi_i | \eta_j \rangle = \langle x_i | y_j \rangle \langle \bar{x}_i | \bar{y}_j \rangle = M_{ij}$, that is, $M = \Xi^\dagger H$, where Ξ and H are the matrices with columns ξ_i and η_j . Hölder's inequality for Schatten norms, $\|\Xi^\dagger H\|_1 \leq \|\Xi\|_2 \|H\|_2$ [7], gives the claim, since

$$\|\Xi\|_2^2 = \sum_i \|\xi_i\|^2 = \sum_i \|x_i\|^4 = \sum_i \langle a_i | L^2 | a_i \rangle^2 = I_a(|X^\dagger|)$$

and

$$\|H\|_2^2 = \sum_j \|y_j\|^4 = \sum_j \langle b_j | RW^\dagger WR | b_j \rangle^2 = \sum_j \langle b_j | R^2 | b_j \rangle^2 = I_b(|X|). \quad \square$$

3.3. Proof of Theorem 5. Let $|\psi\rangle$ be a unit vector with coefficient matrix X , so $\|X\|_2 = 1$. By Lemma 6, applied to the families a and $\bar{b} = \{\bar{b}_j\}$,

$$\|M(|\psi\rangle\langle\psi|)\|_1 \leq I_a(|X^\dagger|)^{1/2} I_{\bar{b}}(|X|)^{1/2}.$$

The family $\bar{b}$ has the same frame constants as b : for Hermitian A we have $I_{\bar{b}}(A) = \sum_j \langle b_j | \bar{A} | b_j \rangle^2 = I_b(\bar{A})$, and $\bar{A}$ is Hermitian with the same trace and the same Hilbert–Schmidt norm as A . The matrices $|X^\dagger|$ and $|X|$ are Hermitian, with $\text{Tr } |X^\dagger| = \text{Tr } |X| = \|X\|_1$ and $\text{Tr } |X^\dagger|^2 = \text{Tr } |X|^2 = \|X\|_2^2 = 1$. Inserting them into (1) gives (2). If ψ has Schmidt rank at most r , then X has at

most r nonzero singular values, and the Cauchy–Schwarz inequality gives $\|X\|_1^2 \leq r\|X\|_2^2 = r$. Since $\alpha \geq \beta$, the right-hand side of (2) is nondecreasing in $\|X\|_1^2$, which proves the bound for pure states. Finally, if $\rho = \sum_t p_t |\psi_t\rangle\langle\psi_t|$ with $p_t \geq 0$, $\sum_t p_t = 1$ and each ψ_t of Schmidt rank at most r , then $M(\rho) = \sum_t p_t M(|\psi_t\rangle\langle\psi_t|)$, and the triangle inequality for $\|\cdot\|_1$ completes the proof. $\square$

4. FRAME CONSTANTS OF MUBS AND OF EQUIANGULAR MEASUREMENTS

We work in the real Hilbert space $\text{Herm}(d)$ of Hermitian matrices with $\langle A, B \rangle = \text{Tr}(AB)$. For a unit vector v , $\langle v|A|v \rangle = \langle P_v, A \rangle$ with $P_v = |v\rangle\langle v|$. Every Hermitian A decomposes orthogonally as $A = \frac{\text{Tr } A}{d} \mathbb{I} + A_0$ with $\text{Tr } A_0 = 0$, and $\|A_0\|_2^2 = \text{Tr}(A^2) - (\text{Tr } A)^2/d$.

Lemma 7 (MUB frames). *The family of all vectors of m mutually unbiased bases has frame constants $(m, 1)$:*

$$\sum_{k=1}^m \sum_{a=1}^d \langle e_a^k | A | e_a^k \rangle^2 \leq \text{Tr}(A^2) + (m-1) \frac{(\text{Tr } A)^2}{d}.$$

Equality holds if and only if $A_0 \in S_1 \oplus \cdots \oplus S_m$, with S_k as in the proof.

Proof. Fix k and let $P_a^k = |e_a^k\rangle\langle e_a^k|$. The P_a^k , $a = 1, \dots, d$, are orthonormal in $\text{Herm}(d)$ and span $V_k = \mathbb{R}\mathbb{I} \oplus S_k$, where $S_k = \{\sum_a c_a P_a^k : \sum_a c_a = 0\}$ is orthogonal to $\mathbb{I}$. Hence

$$\sum_a \langle P_a^k, A \rangle^2 = \|\Pi_{V_k} A\|_2^2 = \frac{(\text{Tr } A)^2}{d} + \|\Pi_{S_k} A_0\|_2^2,$$

where Π denotes orthogonal projection. For $k \neq l$ the subspaces S_k and S_l are orthogonal:

$$\text{Tr} \left[\left(\sum_a c_a P_a^k \right) \left(\sum_b c'_b P_b^l \right) \right] = \sum_{a,b} c_a c'_b |\langle e_a^k | e_b^l \rangle|^2 = \frac{1}{d} \left(\sum_a c_a \right) \left(\sum_b c'_b \right) = 0.$$

By Bessel's inequality, $\sum_k \|\Pi_{S_k} A_0\|_2^2 \leq \|A_0\|_2^2 = \text{Tr}(A^2) - (\text{Tr } A)^2/d$, with equality if and only if $A_0 \in \bigoplus_k S_k$. Summing over k gives $I(A) \leq m(\text{Tr } A)^2/d + \text{Tr}(A^2) - (\text{Tr } A)^2/d$. $\square$

Lemma 8 (EAM frames). *Let $\psi_1, \dots, \psi_n$ be an equiangular measurement in $\mathbb{C}^d$, $n > d$, and $c = c_{n,d}$. For every Hermitian A ,*

$$\sum_{a=1}^n \langle \psi_a | A | \psi_a \rangle^2 \leq \frac{n}{d^2} (\text{Tr } A)^2 + (1-c) \left(\text{Tr}(A^2) - \frac{(\text{Tr } A)^2}{d} \right),$$

with equality if A lies in the real span of the projectors $P_a = |\psi_a\rangle\langle\psi_a|$. Consequently the vectors $\sqrt{d/n} \psi_a$ have frame constants $(\frac{d}{n}, \frac{d(d-1)}{n(n-1)})$.

Proof. Let $T : \mathbb{R}^n \rightarrow \text{Herm}(d)$, $T(x) = \sum_a x_a P_a$, with adjoint $(T^*A)_a = \langle P_a, A \rangle$. Then $\sum_a \langle \psi_a | A | \psi_a \rangle^2 = \|T^*A\|^2 = \langle A, TT^*A \rangle$. The Gram matrix is $T^*T = G = (1-c)\mathbb{I}_n + cJ_n$, where J_n is the all-ones matrix. It has eigenvalue $1-c+cn = n/d$ on the all-ones vector $\mathbf{1}_n$ and eigenvalue $1-c$ on $\mathbf{1}_n^\perp$. Since $TT^*T = TG$:

- $T\mathbf{1}_n = \sum_a P_a = \frac{n}{d}\mathbb{I}$ is an eigenvector of TT^* with eigenvalue n/d ;
- TT^* acts as $1-c$ on $V = T(\mathbf{1}_n^\perp) = \{\sum_a x_a P_a : \sum_a x_a = 0\}$;
- TT^* vanishes on $(\text{range } T)^\perp$.

Every element of V is traceless, so $V \perp \mathbb{I}$ and $TT^* = \frac{n}{d}\Pi_{\mathbb{R}\mathbb{I}} + (1-c)\Pi_V$. Therefore

$$\langle A, TT^*A \rangle = \frac{n}{d} \cdot \frac{(\text{Tr } A)^2}{d} + (1-c) \|\Pi_V A_0\|_2^2 \leq \frac{n}{d^2} (\text{Tr } A)^2 + (1-c) \|A_0\|_2^2.$$

If A lies in range $T = \mathbb{R}\mathbb{I} \oplus V$, then $A_0 \in V$ and equality holds. For the rescaled vectors, multiply by $(d/n)^2$. Using $1 - c = \frac{n(d-1)}{d(n-1)}$,

$$\frac{d^2}{n^2} \left[\frac{n}{d^2} (\text{Tr } A)^2 + (1 - c) \left(\text{Tr } A^2 - \frac{(\text{Tr } A)^2}{d} \right) \right] = \beta \text{Tr}(A^2) + (\alpha - \beta) \frac{(\text{Tr } A)^2}{d},$$

with $\alpha = d/n$ and $\beta = \frac{d(d-1)}{n(n-1)}$, and $\alpha \geq \beta$ because $n > d$. $\square$

Proof of the bounds in Theorems 2 and 3. Apply Theorem 5 with the frame constants of Lemmas 7 and 8. For EAMs, use $a_i = \sqrt{d/n} \psi_i$ and $b_j = \sqrt{d/n} \varphi_j$, so that $M(\rho) = P_n(\rho)$. $\square$

5. TIGHTNESS

Proposition 9. (1) Let $\{e_a^k\}$ be m MUBs in $\mathbb{C}^d$ and $1 \leq r \leq d$. Let Bob use the conjugate bases $f_b^l = e_b^l$ (which are again MUBs). Let Π be the projector onto the span of $e_1^1, \dots, e_r^1$, and let ψ have coefficient matrix $X = \Pi/\sqrt{r}$. Then ψ is a unit vector of Schmidt rank r and $\|Q_m(|\psi\rangle\langle\psi|)\|_{\text{tr}} = 1 + (m-1)r/d$.
 (2) Let $\{\psi_a\}$ be an EAM, let Bob use $\varphi_b = \bar{\psi}_b$, and let Π be a rank- r projector in the real span of the P_a . The state with $X = \Pi/\sqrt{r}$ attains the bound of Theorem 3. Rank- r projectors in this span exist for $r = 1$ (P_a), $r = d$ ($\mathbb{I} = \frac{d}{n} \sum_a P_a$) and $r = d-1$ ($\mathbb{I} - P_a$), and for every r when the P_a span $\text{Herm}(d)$, as for SICs.

Proof. In both cases $M_{ij} = |\langle a_i | X | a_j \rangle|^2$ with $X \succeq 0$, where a is Alice's family (rescaled by $\sqrt{d/n}$ in case 2). With $x_i = X^{1/2} a_i$ we get $M_{ij} = \langle x_i \otimes \bar{x}_i | x_j \otimes \bar{x}_j \rangle$. So M is a Gram matrix, hence positive semidefinite, and $\|M\|_1 = \text{Tr } M = \sum_i \langle a_i | X | a_i \rangle^2 = I_a(X)$.

In case 1, $\langle e_a^1 | \Pi | e_a^1 \rangle \in \{0, 1\}$ with r ones, and $\langle e_a^k | \Pi | e_a^k \rangle = \sum_{b \leq r} |\langle e_a^k | e_b^1 \rangle|^2 = r/d$ for $k \neq 1$. So $I(\Pi/\sqrt{r}) = \frac{1}{r} (r + (m-1)d \frac{r^2}{d^2}) = 1 + \frac{(m-1)r}{d}$.

In case 2, equality holds in Lemma 8 because X lies in the span of the P_a . With $\text{Tr } X = \sqrt{r}$ and $\text{Tr } X^2 = 1$, the value of $I_a(X)$ is the right-hand side of (1) at $(\text{Tr } A)^2 = r$, which is the bound. $\square$

Remark 10 (intermediate Schmidt numbers for equiangular measurements). For the simplex EAM ($n = d+1$) the span of the P_a has dimension $d+1$, and the construction above does not reach $2 \leq r \leq d-2$. Numerical optimisation over pure states of Schmidt rank r gives maxima strictly below the bound in all cases we tried: $0.69496 < 0.7$ for $(d, r) = (4, 2)$; $0.73100 < 0.73333$ and $0.76026 < 0.76667$ for $d = 5, r = 2, 3$; and $0.78230 < 0.78571$ for $(d, r) = (6, 3)$. For $r = d-1$ the bound is attained, as proved above. We do not claim tightness at intermediate r for general EAMs.

6. REMARKS

Remark 11 (a stronger inequality). For a pure state with coefficient matrix X (not necessarily normalised), the proof gives $\|Q_m\|_{\text{tr}} \leq \|X\|_2^2 + (m-1)\|X\|_1^2/d$. Theorem 2 follows from this via $\|X\|_1^2 \leq r\|X\|_2^2$.

Remark 12 (unequal sets and dimensions). The proof of Theorem 5 applies verbatim to $\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}$, with the frame inequality (1) in the respective dimension. For m_A MUBs in $\mathbb{C}^{d_A}$ and m_B MUBs in $\mathbb{C}^{d_B}$ it gives $\|Q\|_{\text{tr}} \leq [(1 + (m_A - 1)r/d_A)(1 + (m_B - 1)r/d_B)]^{1/2}$ for $\text{SN}(\rho) \leq r$.

Remark 13 (comparison with the triangle inequality). Write a unit vector of Schmidt rank r as $\psi = \sum_s \sqrt{\lambda_s} u_s \otimes v_s$. Expanding $Q_m(|\psi\rangle\langle\psi|)$ in the Schmidt terms and using the triangle inequality leads to bounds of the form $\|X\|_1^2 + (m-1)\|X\|_2^2/d$, that is, $r + (m-1)/d$ in the worst case. This exceeds $1 + (m-1)r/d$ by $(r-1)(d+1-m)/d$, which is positive for $r \geq 2$ and

$m \leq d$. The Gram factorisation of Lemma 6 places the trace norm $\|X\|_1^2$ on the “coherent” term $(m-1)/d$ and the Hilbert–Schmidt norm on the diagonal term, which is the tight arrangement.

Remark 14 (scope). Conjecture 1 of [1], on complete sets of MUBs and SICs, is not addressed here.

7. FORMAL VERIFICATION

Theorems 2 and 3 are formalised in Lean 4 (version 4.33.1) [8] with the Mathlib library [9] (revision 0df444a3). The formal statements are:

```
theorem tavakoli_morelli_conjecture_3 (hm : 1 ≤ m) (e f : Fin m → Fin d → Fin d → ℂ)
  (he : IsMUB d m e) (hf : IsMUB d m f) (ρ : Matrix (Fin d × Fin d) (Fin d × Fin d) ℂ)
  (hρ : HasSchmidtNumberLE ρ r) :
  traceNorm (probMatrix e f ρ) ≤ 1 + ((m : ℝ) - 1) * r / d

theorem tavakoli_morelli_conjecture_2 (hdn : d < n) {ψ φ : Fin n → Fin d → ℂ}
  (hψ : IsEAM d n ψ) (hφ : IsEAM d n φ) (ρ : Matrix (Fin d × Fin d) (Fin d × Fin d) ℂ)
  (hρ : HasSchmidtNumberLE ρ r) :
  traceNorm (eamProbMatrix ψ φ ρ) ≤ ((d : ℝ) * (d - 1) + r * (n - d)) / (n * (n - 1))
```

The definitions are as follows.

- `traceNorm` is the sum of the singular values, defined through the eigenvalues of $Q^T Q$.
- `HasSchmidtNumberLE` states that ρ is a finite convex combination of projectors onto unit vectors of Schmidt rank at most r .
- `IsMUB` and `IsEAM` are the definitions of Section 2.
- `probMatrix` and `eamProbMatrix` are the matrices Q_m and P_n .

The formal proof proceeds through the variational characterisation of the trace norm. It shows, using a singular value decomposition built from Mathlib’s spectral theorem, that $\|Q\|_{\text{tr}} \leq \gamma$ as soon as $\text{Tr}(O^T Q) \leq \gamma$ for every real orthogonal O . The bound on $\text{Tr}(O^T Q)$ (theorems `tavakoli_morelli_conjecture_3_dual` and `_2_dual`) is proved along the lines of Lemma 6 and Lemmas 7 and 8.

All main theorems depend only on Lean’s standard axioms (`propext`, `Classical.choice`, `Quot.sound`). The development contains no `sorry` and no `native_decide`. As sanity checks the repository also contains non-vacuity examples (qubit MUBs and a Bell state).

DATA AVAILABILITY

The Lean project, the numerical checks and the verification logs are available at <https://github.com/anshM123/Tavakoli-Morelli-Conjecture>.

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